

Make An Arduino-Based Wireless Frequency Meter

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This project describes an Arduino-based wireless frequency meter designed to measure frequency of sinusoidal AC signals in the range of 50Hz to 3kHz. The authors' prototypes of the transmitter and receiver sides are shown in Figs 1 and 2, respectively.

The block diagrams of the transmitter and receiver sides are shown in Figs 3 and 4, respectively.

Circuit and working

The main components used in this project and their role is described below.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and fourteen digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI and I2C.

The MCU can operate at a clock frequency of 16MHz. In our project, Arduino Uno is used on the receiver side. Digital I/O pin 5 is used as input pin and is connected to DATA pin of 434MHz RX module. Digital I/O pins 7, 8, 9, 10, 11 and 12 of Arduino are used to interface with the 16x2 character LCD to display the value of the measured frequency.

Rectifier diodes. Rectifier diodes (1N4007) are used to rectify the alternating input signal to pulsating DC on transmitter side.

MCT2E. MCT2E opto-coupler IC is used to isolate input side from the next stage and also produce signals from pulsating DC waveform on transmitter side.

LCD. The 16x2 LCD, present at the receiver end, is used to display the value of frequency. It is interfaced with Arduino Uno board in four-bit

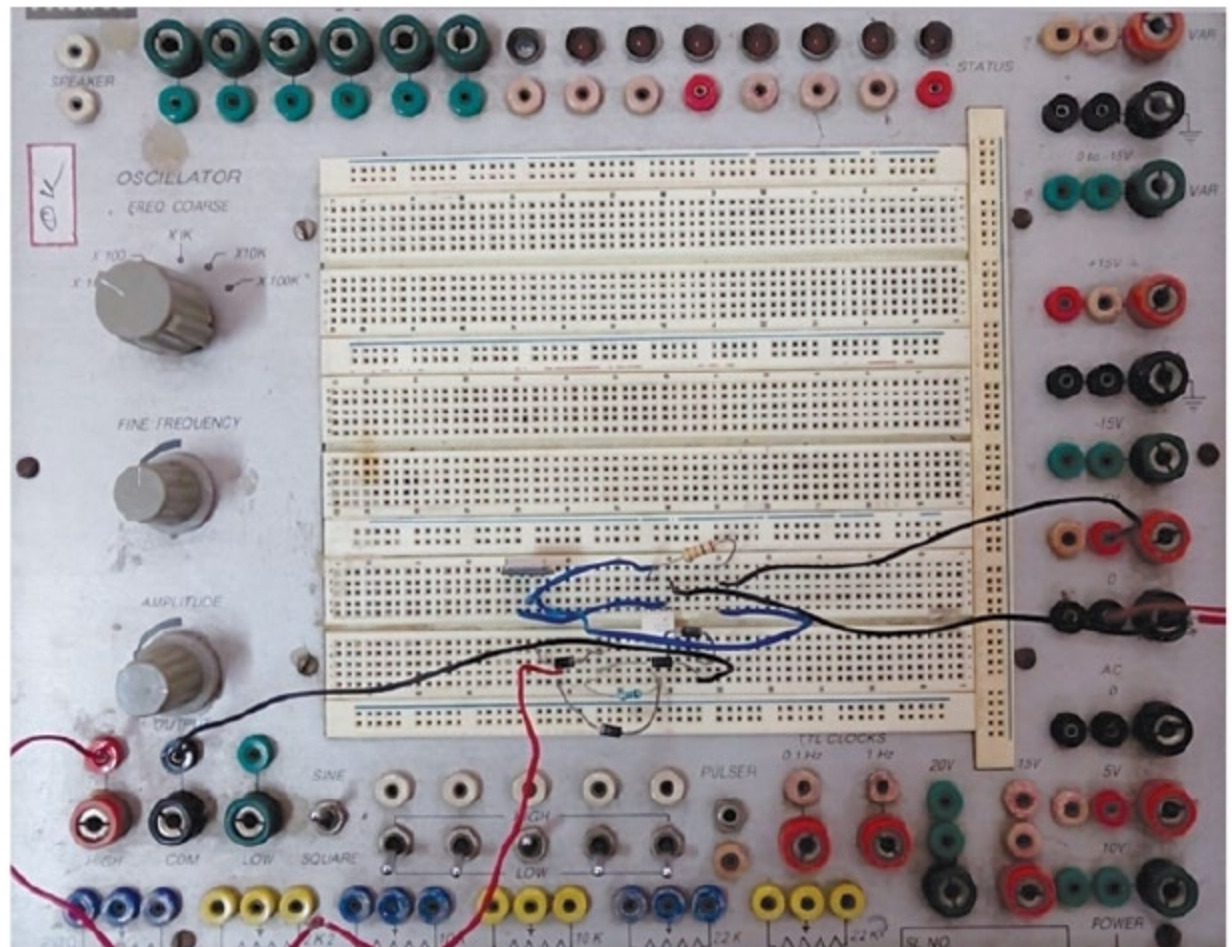


Fig. 1: Authors' prototype of the transmitter side

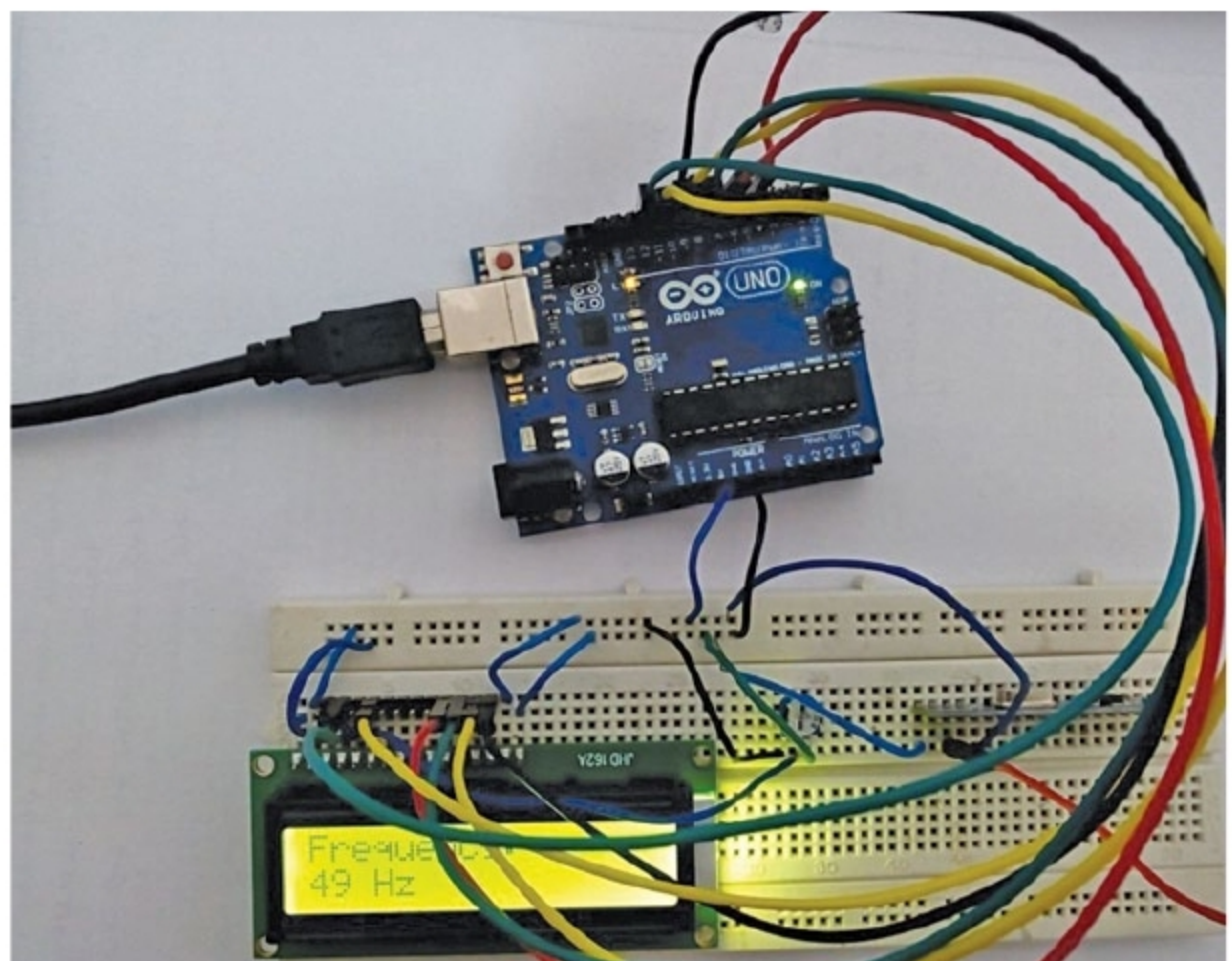


Fig. 2: Authors' prototype of the receiver side